Python & R Programming Self-Learning Questions

Topic: Python Basics & Control Flow

1. Explore why Python is widely used as a glue language in Big Data ecosystems.

2. Study how control flow constructs translate algorithmic logic into executable programs.

3. Observe and document how improper loop and conditional design leads to inefficiency in large-scale data processing.

Topic: Strings & Processing

1. Analyze why string manipulation is central to data cleaning and preprocessing tasks.

2. Study immutability of strings and tuples and its impact on memory and performance.

3. Research and study real-world text processing problems such as normalization, validation, and pattern detection.

Topic: Dictionaries & Data Models

1. Explore why key–value data structures dominate Big Data storage and processing models.

2. Study dictionary comprehension as a bridge between procedural logic and declarative data transformation.

Topic: Lists & Scalability

1. Implement and check how list operations scale poorly with large datasets.

2. Study when lists are useful for prototyping versus when distributed data structures are required.

Topic: Functions & Modularity

1. Study how modular programming improves reusability and maintainability in data pipelines.

2. Explore why functions are first-class citizens in Python and how this enables functional programming paradigms.

3. Examine and document pickling as a mechanism for object serialization and its relevance in data transfer and storage.

Topic: Tuples & Fixed Records

1. Analyze why tuples are commonly used to represent fixed-schema records.

2. Study tuple-based data modeling as a precursor to rows in DataFrames and RDDs.

Topic: OOP, Generators & Decorators

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1. Explore how object-oriented design improves scalability and extensibility of software systems.

2. Study generators as a memory-efficient way to process large sequences.

3. Observe and document how decorators enable cross-cutting concerns such as logging and monitoring.

Topic: Exception Handling & Logging

1. Study why exception handling is critical in long-running data processing jobs.

2. Explore logging as an operational necessity rather than a debugging convenience.

Topic: NumPy & Pandas Processing

1. Study why vectorized operations outperform iterative logic in data processing.

2. Explore Pandas as a local-scale analytics tool and its limitations for Big Data.

3. Observe and analyse how data cleaning decisions affect downstream analytics and models.

Topic: Web & Visualization

1. Examine ethical and technical considerations in data collection from the web.

2. Study visualization as a tool for insight validation rather than presentation.

Topic: Environment & Media Data

1. Study why isolated environments are essential in complex data ecosystems.

2. Observe how non-tabular data (images, audio) introduces new data handling challenges.

Topic: Database Integration

1. Reflect on challenges of integrating transactional databases with analytics workflows.

2. Study Python's role as an integration layer between databases and processing engines.

Topic: Introduction to R

1. Reflect on why R emerged as a dominant language for statistical computing and analytics rather than general-purpose programming.

2. Study the difference between using R as a calculator versus using R as an analytical environment.

3. Explore how IDEs such as RStudio influence productivity, reproducibility, and workflow design.

Topic: R Data Structures

1. Analyze how R's data structures (vectors, lists, data frames) are designed around statistical thinking.

2. Study the role of packages in extending R's capabilities and how dependency management works.

3. Reflect on challenges involved in importing data from heterogeneous sources such as text files, Excel, and databases.

Topic: Functions & Reporting

1. Study how writing functions improves reusability and clarity in analytical code.

2. Reflect on the importance of reproducible research and interactive reporting using R Markdown.

3. Analyze how exploratory analysis and visualization support evidence-based conclusions in case studies.